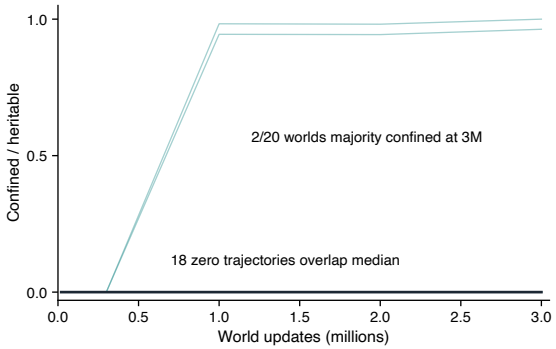
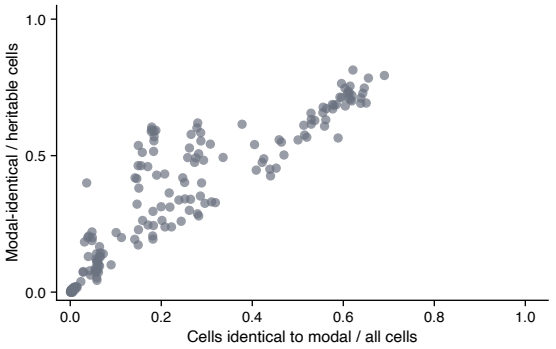
a Random-register reset: Z80 L16, 20 worlds



Thin = worlds; black median; cells confined / heritable cells
Confined: no partner instruction-stream fetch in 16/16 traces

b Random cells versus the modal tape



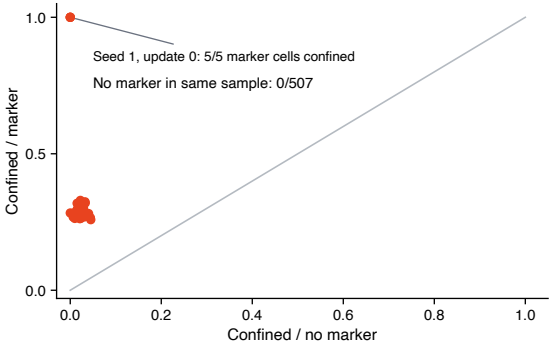
One dot = final world sample; 170 worlds in cohorts of c
Y: sampled heritable cells identical to modal / sampled heritable cells

c Class denominators by cohort

Cohort	Worlds	Cell rows	Conf. rows	T rows
G, L16	20	3938	2624	259
G, L20	20	3757	2032	358
G, L50	20	5152	1165	19
G, L64	20	5444	897	320
I, L16	10	1217	1217	1008
K, L32	20	5519	741	149
L, L16	10	3210	2649	112
L, L20	10	2556	1783	253
M, L16	20	3966	2536	0
M, L32	20	5600	11	0

Pooled sampled-cell rows; confined and T are subsets
G benign Z80; I lethal Z80; K aligned L32; L 10M extensions
M 8080 subset; L in labels = tape bytes (not cohort L)

d Sequence marker is not confinement



Z80 L50 matched-pair invasion: 3 worlds, 24 samples
20 F0 marker; axes: confinement among positive/negative cells
512 cells/sample; equal-fraction diagonal

e Exploratory detection order: any detection

Cohort	Worlds	R<T	Same	T<R	R only	T only	None
G, L16	20	0	1	1	17	1	0
G, L20	20	4	9	0	3	3	1
G, L50	20	8	1	0	11	0	0
G, L64	20	18	0	0	2	0	0
I, L16	10	1	1	5	0	3	0
K, L32	20	1	8	2	5	0	4
L, L16	10	0	0	1	9	0	0
L, L20	10	4	4	2	0	0	0
M, L16	20	0	0	0	20	0	0
M, L32	20	0	0	0	8	0	12

R = confined ≤2 scored sites; T = confined ≥5 scored sites
R<T: R observed first; Same: same sampled update; T<R: T first
Any = positive estimated class share; n = worlds; not ancestry

f Detection-order threshold sensitivity

Threshold	R<T	Same	T<R	Both	n
G, L16 >50%	0	0	0	0	20
G, L16 >10%	0	0	1	1	20
G, L20 >50%	0	0	2	2	20
G, L20 >10%	6	5	0	11	20
G, L50 >50%	0	0	0	0	20
G, L50 >10%	1	0	0	1	20
G, L64 >50%	2	0	2	4	20
G, L64 >10%	8	0	0	8	20
I, L16 >50%	1	0	0	1	10
I, L16 >10%	1	1	1	3	10
K, L32 >50%	0	0	1	1	20
K, L32 >10%	1	5	3	9	20
L, L16 >50%	0	0	1	1	10
L, L16 >10%	0	0	1	1	10
L, L20 >50%	0	0	2	2	10
L, L20 >10%	5	3	2	10	10
M, L16 >50%	0	0	0	0	20
M, L16 >10%	0	0	0	0	20
M, L32 >50%	0	0	0	0	20
M, L32 >10%	0	0	0	0	20

R/T and order keys as e; threshold = class share / heritable cells
Both detected = sum of the three order columns; n = worlds
Site: ≥half tested byte values heritable, allele in ≥50% qualifying copies